

ZIAD ASHRAF

Software engineer (MERN)

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TECHNICAL SKILLS

- **Backend Engineering:** Node.js, Express.js, RESTful API Design
- **Frontend Engineering:** React.js, Tailwind CSS, Progressive Web Apps (PWA)
- **Databases & BaaS:** MongoDB, PostgreSQL, Firebase, Supabase
- **Security & DevSecOps:** Proficient in mitigating OWASP Top 10 vulnerabilities and implementing secure-by-design architectures
- **Languages:** C++, JavaScript, C#
- **DevOps & Infrastructure:** Docker (Containerization), Linux Environment Administration.
- **Version Control & Collaboration:** Git, GitHub (Branching, PR Reviews, Git Workflows).
- **Core Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Competitive Programming

EDUCATION

Bachelor of Science in Computers and Artificial Intelligence 2025 – Present (Expected 2029)

South Valley National University

Relevant Coursework: Data Structures, Analysis and Design of Algorithms, Operating Systems, Databases, Object-Oriented Programming (OOP), Web Security

PROFESSIONAL PROJECTS

Dreksion — IEEE Award-Winning Driving School Platform

React.js, Node.js, Express.js, MongoDB, Tailwind CSS

- Architected and engineered a secure, full-stack driving school management platform, winning a competitive software project award affiliated with the IEEE.
- Implemented robust session management and secure user authentication/authorization flows to protect user accounts and sensitive booking data.
- Hardened API endpoints against unauthorized access, ensuring data integrity across the client-server communication layer.

Educational Platform (MERN Stack)

Node.js, Express.js, MongoDB, Custom REST APIs

- Built a robust, scalable server-side backend supporting core educational platform workflows using custom REST APIs.
- Resolved critical Node.js version compatibility and dependency conflicts across Mongoose and Express, stabilizing the development pipeline.
- Designed strict schema-level validation rules to prevent malicious input injection and enforce data consistency.
- Secured environment configuration management to protect sensitive credentials and prevent accidental exposure of secrets.

Secured E-Commerce Platform

Node.js, Express, MongoDB, Payment Gateway Integrations

- Engineered an end-to-end secure checkout pipeline, integrating payment gateway services while safeguarding transaction integrity.
- Diagnosed and resolved critical backend race conditions affecting order processing under concurrent load.
- Implemented rate limiting middleware to defend against API abuse, brute-force attempts, and fake/automated requests.
- Secured payment webhook endpoints against spoofing and replay attacks, ensuring reliable transaction confirmation.

Qena University Application Portal

Full-Stack Integration, Database Management

- Built a deployment-ready application portal responsible for handling sensitive student registry data.
- Designed and enforced strict Role-Based Access Control (RBAC) to eliminate privilege escalation risks across user roles.
- Conducted security audits on all input forms, identifying and remediating Cross-Site Scripting (XSS) vulnerabilities.
- Maintained database integrity and consistency throughout high-volume student registration cycles.